



From the Manuscript to the Digital Scholarly Edition

Hannah Busch

Cologne Center for
eHumanities (CCeH)

SunoikisisDC - Digital Classics Summer 2026
May 5 2026

Schedule

1. Collections & Editions
2. From Parchment to Cyberspace
3. Imaging Techniques
4. Interfaces
5. Metadata
6. Working with Digitized Manuscripts

Collections/Virtual Manuscript Libraries



e-codices



SINAI MANUSCRIPTS DIGITAL LIBRARY
A publication of St. Catherine's Monastery of the Sinai, Egypt

Bibliotheca Palatina – digital



eCodicesNL
the virtual reading room



UNIVERSITÄT
ZU KÖLN

Cologne Papyrus Collection

**KÖLNER PAPYRI**

Sammlung
Publikationen
Projekte
About

Login Inv.-Nr., Suchbegriff...

Herzlich willkommen bei der Kölner Papyrussammlung!

Inv.-Nr., Suchbegriff... Sammlung durchsuchen

Die Webseite bietet einen recherchierbaren Katalog der Bestände der Kölner Papyrussammlung und Zugang zu Abbildungen sowie den wichtigsten Metadaten zu den einzelnen Objekten. Für Ihre Recherche nutzen Sie auch die facetierte Sammlungsübersicht sowie das klassische, komplexe Suchformular.



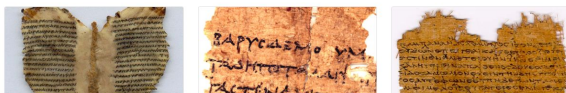
4482 Objekte

4902 Texte

105 Publikationsbände

1357 Edierte Texte

Ausgewählte Stücke der Sammlung



Kölner Papyri / Sammlungsübersicht / Suchergebnisse

Sammlungsübersicht

Objektfiler

4482 Sammlungsobjekte gefunden

Freie Suche

Inv.-Nr., Suchbegriff...

Suchoptionen

☐ Nur Ergebnisse mit Bildern

☐ Nur Ergebnisse mit identifizierten Texten

☐ Nur Ergebnisse mit Transkriptionen

Sortieren nach

Inventarnr.

aufsteigend

Anwenden

Material

Papyrus 4399

Carbonisiert 247

1 2 3 ... 150 1 (Inv. 00001 ... Inv. 00038)



Inv. 00001, Köln, Papyrussammlung

Recto

6. Jh.



Inv. 00002, Köln, Papyrussammlung

Recto: Adespotum: Prosafragment

2. Jh.



Inv. 00003, Köln, Papyrussammlung

Recto und Verso: Aristophanes, Lysistrata 145-153: 182-199

4. Jh.



Inv. 00004, Köln, Papyrussammlung

Recto und Verso

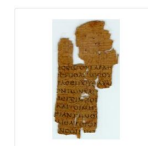
Transkription

Quelle: Papyri.info

1 [-ca.?-] ἡρώδης
γενέσκειν αὐτὸν βασιλεῖα
παύσειν, ὅτι τὸ ... I ... V
ἀποκρίνεται καὶ οὐκ ἔλαβεν αὐτὸν.
5 ἔσπευσά σοι οὐδ' ἐδήλωσά σοι
διὰ τοῦτον μοι τῶν γραμμάτων
τὴν λ' (ἀρουρῶν) δέμα α.
τὴν κη (ἀρουρῶν) τὴν α-
τὸ δέ δέμα διαγράφεται
10 τολάντων ἂ καὶ κη
ἡ δὲ τὴν διαγράφεται
(τολάντων) μετ' ἡ (τολάντων) μετ' ἡ
οὐν πρὸς τὸν M ...]ου
ἀποκρίνεται [-ca.?-]
15 διὰ τοῦ α [-ca.?-]
ταυ[-ca.?-]
...
V
κυρία μοι πατρί
Θεογένετι
Apparatus
r.2. l. γενέσκειν

Kölner Papyri / Sammlungsübersicht / Suchergebnisse (Seite 1) / Inv. 00002

Inv. 00002, Köln, Papyrussammlung



2R

Q 1422a2554 (jpg)



P Inv. 00002r

Q 1422a2554 (jpg)

© Institut für Altertumswissenschaften an der

Trismegistos-Nummer	63496
Objektyp	Rolle
Material	Papyrus
Dimensionen	Breite: 5,6 cm • Höhe: 10,2 cm
Texte	Recto: Adespotum: Prosafragment
Bibliographie	Ausblung P.Köln III Tafel Va

Texte

Recto: Adespotum: Prosafragment	
Textsorte	Datierung
literarisch	2. Jh.
Herkunft	

<https://papyri.uni-koeln.de/>

THE ARCHIMEDES PALIMPSEST

[PALIMPSEST](#)

[IMAGING](#)

[SCHOLARSHIP](#)

[DIGITAL](#)

[SLIDESHOW](#)

[LINKS](#)



Welscher Gast digital



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How?

Digitization processes and technologies



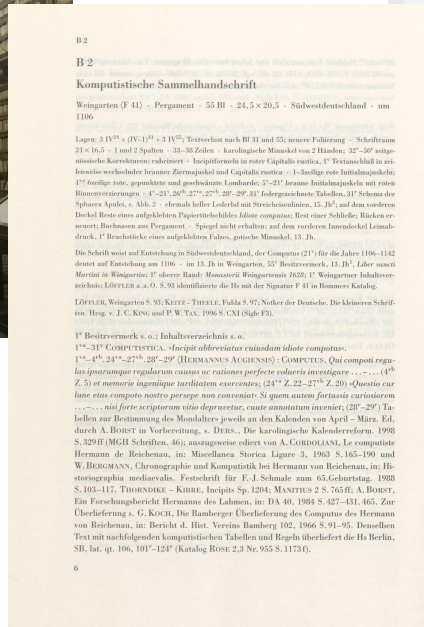
From Parchment to Cyberspace

From the material book to digital representation. A short overview.



From Parchment to Cyberspace



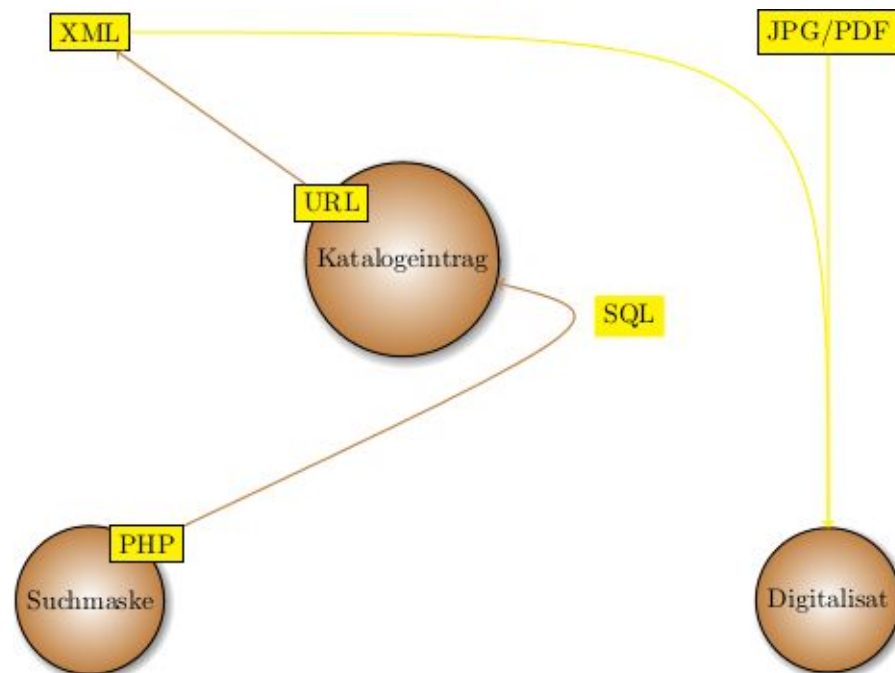


The printed catalogue





Photographic reproduction



Processing and publication

e-codices - Virtual Manuscript Library of Switzerland

The goal of e-codices is to provide free access to all medieval and a selection of modern manuscripts of Switzerland by means of a virtual library.

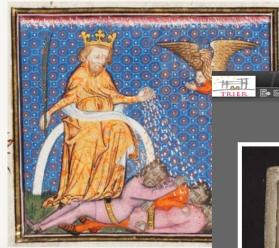
At the moment, the virtual library contains **2959** manuscripts from **100** different collections. The virtual library will be continuously updated and extended.

Collections

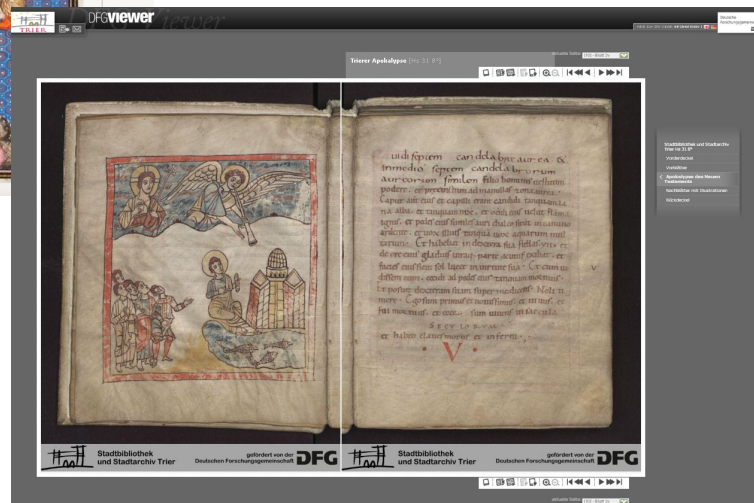
Swiss Collections

Location, Library / Collection	Documents
All Libraries and Collections	2959
Aarau, Aargauer Kantonsbibliothek	43
Aarau, Staatsarchiv Aargau	16
Appenzell, Landesarchiv Appenzell Innerrhoden	36
Basel, Pharmaziemuseum der Universität Basel	10
Basel, Staatsarchiv Basel-Stadt	2
Basel, Universitätsbibliothek	311
Bern, Bürgerbibliothek	235

Exhibition

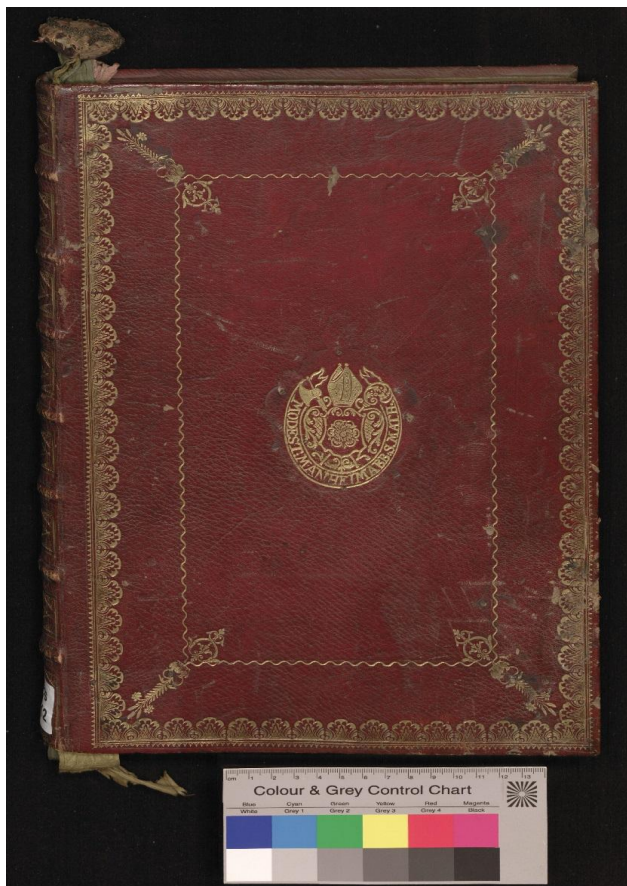


Virtual library and image viewer



The Digital Facsimile

- The Reproduction is not the Original:
 - Proportions
 - Color fastness
 - Dimensionality
 - Artifacts, Noise, Error
- Antidotes
 - Color checker and scale
 - Calculation of systematic deviations



Bibliothek des
Bischöflichen Priesterseminars Trier

gefördert von der
Deutschen Forschungsgemeinschaft

DFG



Stadtbibliothek
und Stadtarchiv Trier

gefördert von der
Deutschen Forschungsgemeinschaft

DFG



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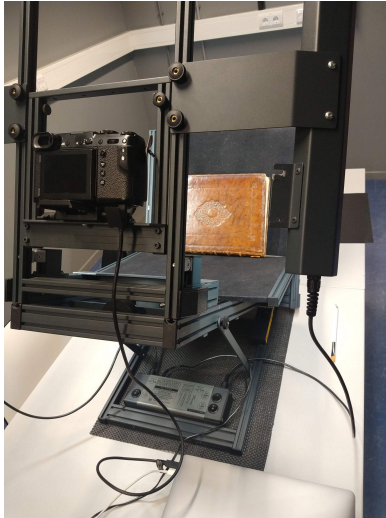
And sometimes you have to be creative...

Disclaimer: Do not imitate,
please! ;)

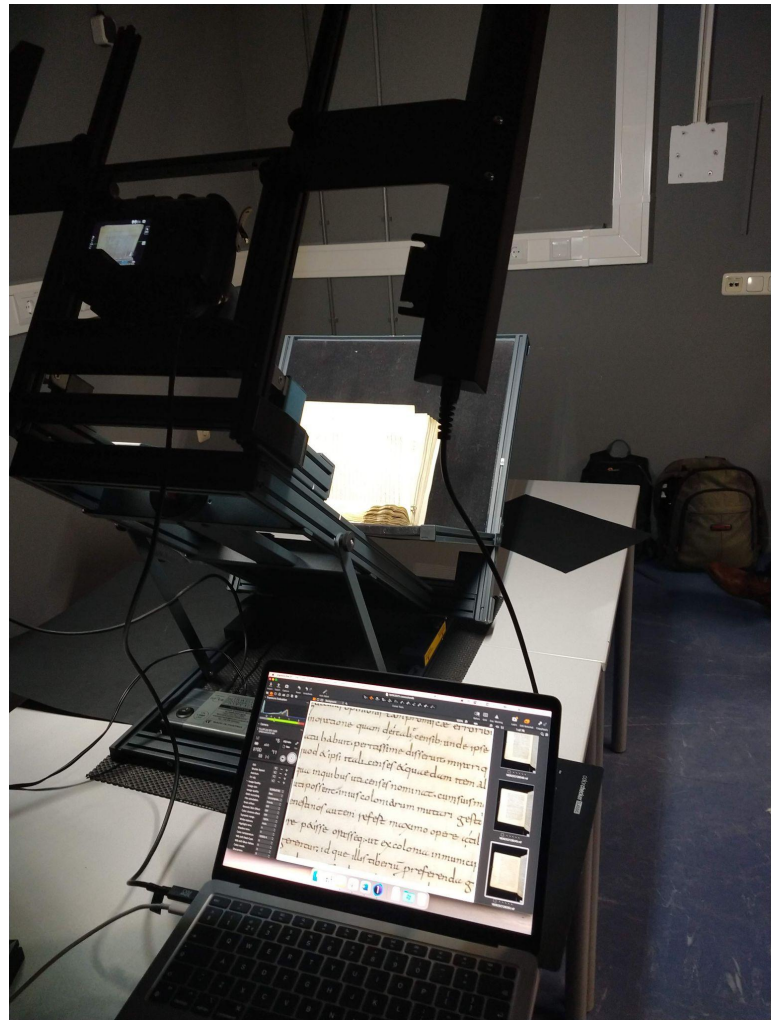


Imaging Techniques

Digital Photography



Digitizing medieval manuscripts with the traveller at Tresoar, Leeuwarden (NL), 2022.



Wolfenbütteler Buchspiegel

Mirror

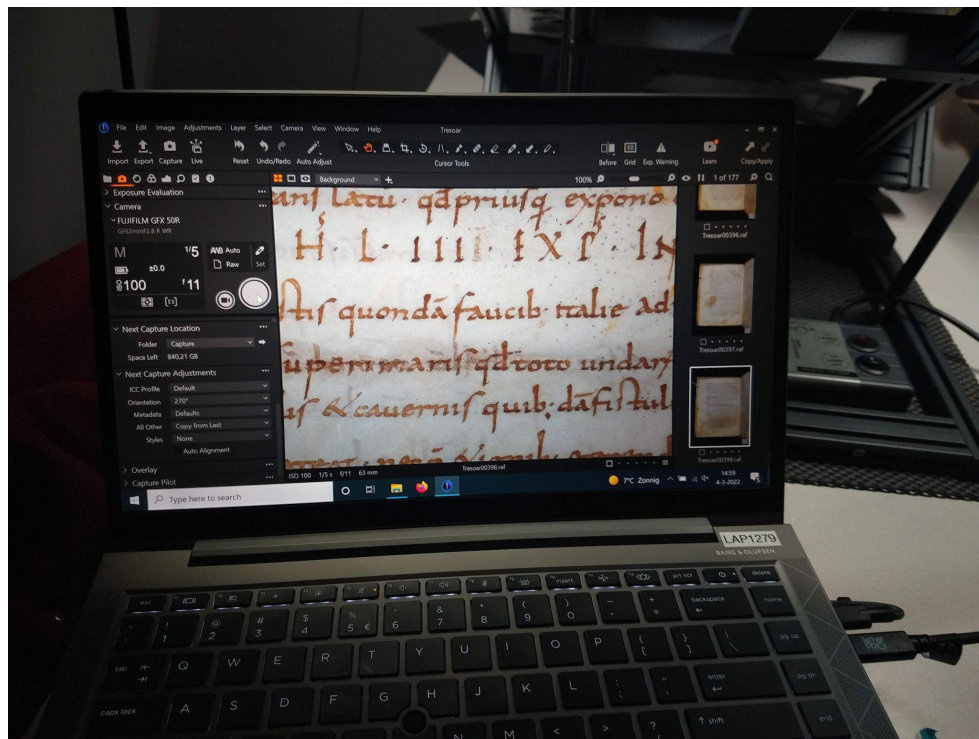
Camera: Shot of single pages via mirror



Bookend:
Less than
45 degree
opening angle



How to Digitize a Manuscript?



Checklist manuscript digitization:

Collection and signature:

Number folia:

Number flyleaf:

Datum:

Before:

- ☐ Colorimeter
- ☐ Colorimeter page/leaf
- ☐ Colorimeter cover
- ☐ Colorimeter back
- ☐ Colorimeter open view

- ☐ Edge bottom

Extra:

- ☐ Open view
- ☐ Fotos without acid-free paper
- ☐ Extras

Notes:

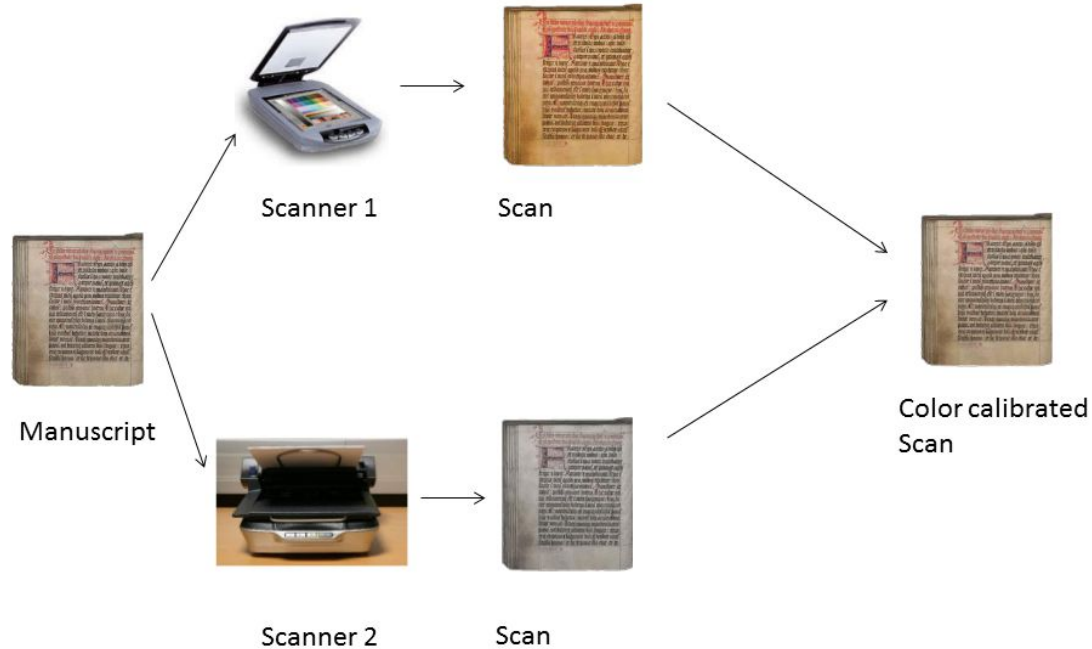
Bookblock:

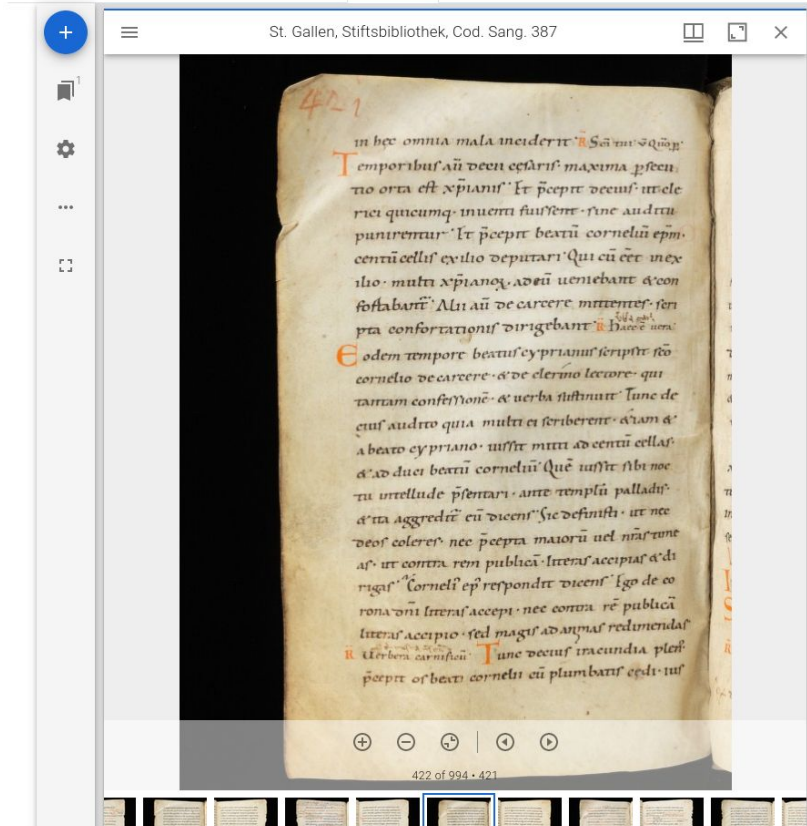
- ☐ Rectos
- ☐ Versos
- ☐ Flyleaves front
- ☐ Flyleaves back
- ☐ Endpaper front
- ☐ Endpaper back

Cover:

- ☐ Front cover
- ☐ Front cover with lineal
- ☐ Back cover
- ☐ Back
- ☐ Edge top

Post-processing





showing fields

uri ⓘ

<https://www.e-codices.unifr.ch/loris/csq/csq-0387/csq-l>

Dataset uri 

<https://www.e-codices.unifr.ch/metadata/iiif/csq-0387/r>

Dataset ID Publisher

e-codices - Virtual Manuscript Library of Switzerland

Repository

Stiftsbibliothek

Shelfmark

Cod. Sang. 387

origPlace ⓘ

origDate ⓘ

Normalized date "notBefore" ⓘ

800

Normalized date "notAfter"

899

Handnote script

Script type as in description ⓘ

Metadata from IIF

Location: St. Gallen

Collection Name: Stiftsbibliothek

Shelfmark: Cod. Sang. 387

Document Type: Manuscript

DOI: 10.5076/e-codices-csg-0387

Title (English): Breviary, summer portion (with neumes)

Material: Parchment

Place of Origin (English): St. Gall

Date of Origin (English): second quarter of the 11th century (additions through the 14th century)

Number of Pages: 984

Dimensions: 24.5-25 x 16-16.5 cm

Online Since: 2010-11-04

Summary (English): Summer portion (Holy Saturday through the end of the church year) of a breviary written at the Abbey of St. Gall between 1022 and 1047 (with readings, prayers, extracts from homilies, antiphons, responses and hymns for the monastic liturgy of the hours), includes additions made as late as the 14th century. The sung sections include neumes. Preceding materials include a fragment of a collections of homilies, a calendar, and computistical texts and tables. The corresponding winter portion of this breviary is found in Cod. Sang. 413. It is among the oldest surviving breviaries produced at St. Gall.

Persons: Translator: Avitus, Alcimus Ecdicius; Author: Lucianus, de Caphargamala

Century: 11th century

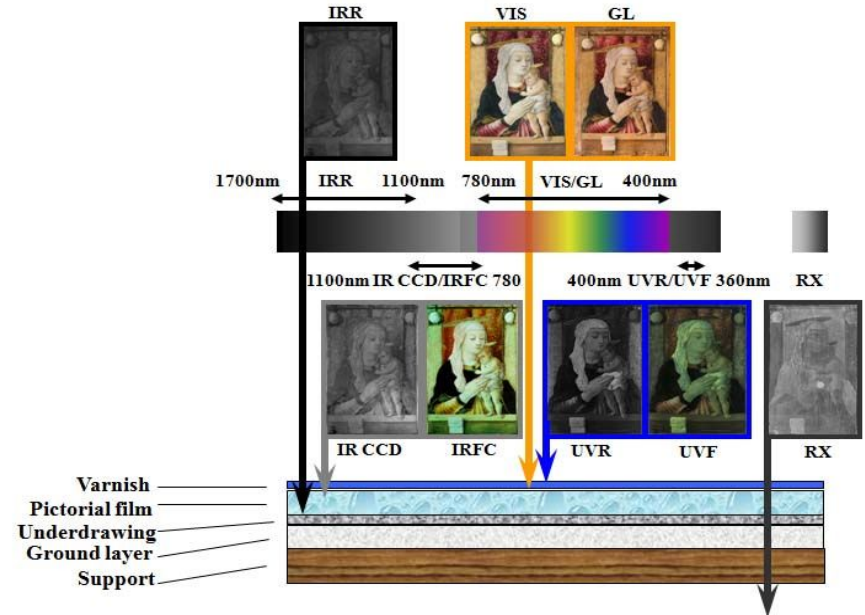
Liturgica christiana: Breviary

Text Language: Latin

uri: https://www.e-codices.unifr.ch/loris/csq/csq-0387/csq-0387_e001.jp2

Dataset uri: <https://www.e-codices.unifr.ch/metadata/iiif/csg-0387/manifest.json>

Multispectral Imaging



Source: Quelle: <http://www.webexhibits.org/pigments/intro/visible.html>

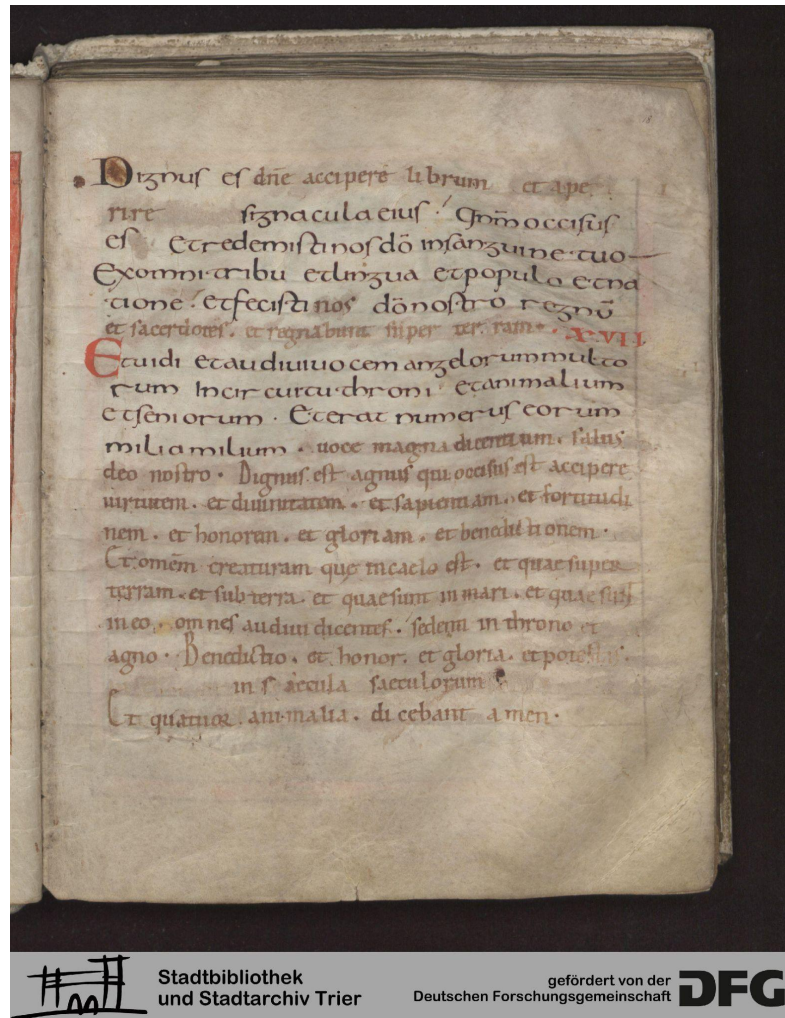
Recovering Lost Texts

- Recovery of under-text in a palimpsest
- faded text, erasures
- damages caused by aging, floods, fires

Palimpsest?

StB. Trier, Hs. 31

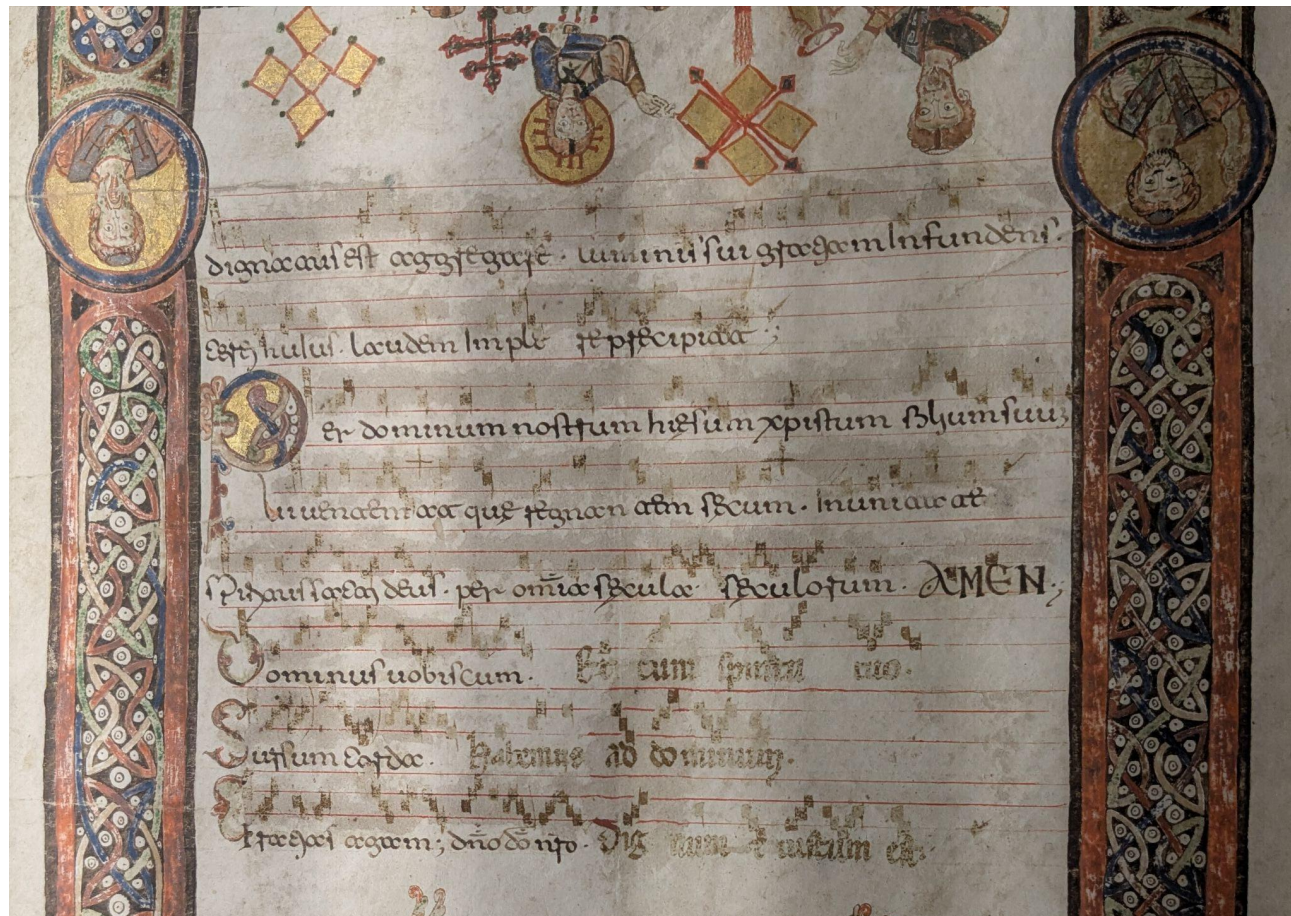
- 9th century
- parchment
- text erased and overwritten



source: stmatthias.uni-trier.de

Any thoughts?

What do you see?

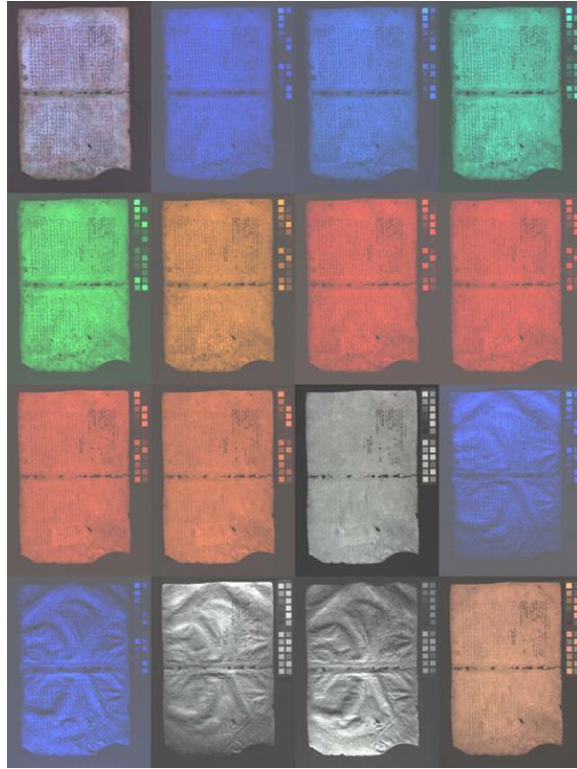


Roll of the Exultet,
Artecclesiae Bari-Bitonto

How does MSI work?

- a multispectral imaging system consists of
 - camera
 - lights
 - special lens
- image processing software
- different things become visible in light with different wavelengths
- non-invasive method
- time consuming!

The Archimedes Palimpsest

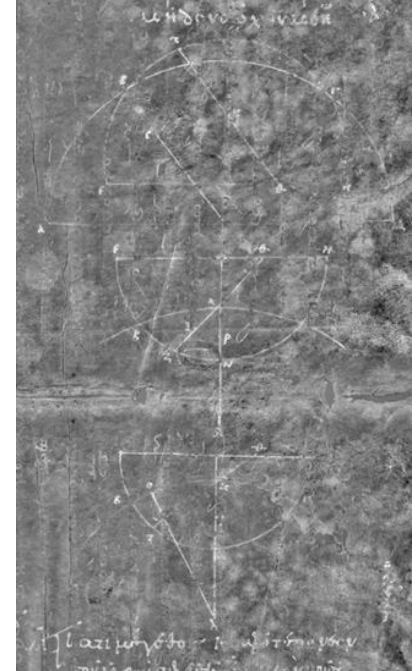


→ why pre-/post-processing matters:

before:



after:



"I did not expect to see a King riding an ostrich on a mountain in North Africa."
Vercelli Mappa Mundi



Interfaces

What users see and interact with

Faceted Browsing

Greek Keyboard | Reset Search

Date 400 – 1500

Filters

Object type

Material

Language

Greek (110)

Latin (2)

Personal names

Family names

Gender

female (1)

male (106)

Milieu

aristocracy (13)

civil (37)

military (17)

monastic (5)

private (30)

secular church (8)

Place names

Aegean Sea (1)

Anatoliki (2)

Armenia (1)

Robert Feind Collection

Seal of N. imperial protospatharios epi tou chrysotriklinou an Cephalonia and Nikopolist1

Seal of Basileios imperial protospatharios and genikos logoth

Seal of Bardas patrikos, imperial protospatharios, and strate

Seal of Manouel Mandromenos

Seal of Theophanios imperial protospatharios and strategos c

Seal of Nikolaos imperial kandidatos and epi tou oikistikou

Seal of Nikephoros patrikos, imperial protospatharios, and st

Seal of Theodosios imperial spatharios and epi ton oikeiakon

Seal of N. imperial spatharokandidatos and asekreitis

Seal of Damianos

Seal of Bardas magistros, anthypatos, patrikos, imperial prot

domestikos of the Scholai

Seal of Theodoros

Seal of Nikolaos imperial spatharios and asekreitis

Seal of Symeon imperial protospatharios and kourator of...

Seal of Nikolaos archon of Demetrias

Seal of Basileios kourator of the strategos

Seal of Maria patrikia

Seal of Martinos strateiates

Seal of Athenaios ilioustrios

Seal of Konstantinos protospatharios, imperial notarios of the

krites epi tou hippodromou and of Macedonia

Seal of Basileios Mauros, proedros and krites epi tou hippod

Seal of Ioannikios monachos and episkopos of Eumenelas

Seal of Maurianos apo hypaton

Seal of Bonosos

<https://feind.sigidoc.cceh.uni-koeln.de/en/>



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→ [Letter](#)

THE LEON LEVY
Dead Sea
Scrolls
DIGITAL LIBRARY

[HOME](#)
[FEATURED SCROLLS](#)
[EXPLORE THE ARCHIVE](#)
[LEARN ABOUT THE SCROLLS](#)
[ABOUT THE PROJECT](#)

Browsing
Manuscripts

IN:
Wadi Murabba'at

ADD FILTERS:

MATERIAL
Papyrus (114)
Parchment (38)
Paper (7)

SCRIPT LANGUAGE
Greek (70)
Hebrew (23)
Hebrew (18)
More options...

COMPOSITION TYPE
Other (84)
Unidentified Text (56)
Bar Kokhba letters (11)
More options...

PERIOD
Roman (107)
Hellenistic-Roman (35)
Islamic (6)

Showing 12 of 159
Results

MUR Numbers
MUR1
18 IMAGES

MUR Deuteronomy
MUR2
8 IMAGES

MUR Mezuzah
MUR5
1 IMAGES

MUR papAccount of
Cereals and Vegetables

<https://www.deadseascrolls.org.il/explore-the-archive>

512 Ergebnisse

ERGEBNISTYP

- ☐ Brief (erfasst) (512)
☒ Brief (transkribiert) (512)
☐ Person (0)

SUCHE ANPASSEN

- ☐ Unschärfe Suche
☐ Alle Wörter (UND)
☐ Beliebige Wörter (ODER)

VERFASSER

+ Haeckel, Karl (512)
Haeckel, Charlotte (14)
Haeckel, Carl Gottlob (9)
Haeckel, Hermine (6)
Haeckel, Heinrich Friedrich Ernst (5)
Sethe, Wilhelmine (2)
Sethe, Bertha (2)
Hahn, Anna Charlotte Wilhelmine (1)
Haeckel, Siegfried (Friedel) Otto Richard (1)
Haeckel, Anna (1)

[Filtern](#)

EMPFÄNGER

Haeckel, Ernst (495)
Haeckel, Carl Gottlob (12)
Haeckel, Charlotte (11)
Haeckel, Agnes (2)
Haeckel, Karl (1)
Gegenbaur, Carl (1)
Wieck, Sophie Marianne (1)
Paetel, Hermann (1)
Paetel, Elwin (1)
Focke, Wilhelm Olbers (1)

[Filtern](#)

ENTSTEHUNGSLAND

Deutschland (446)
Polen (20)
Italien (3)
Schweiz (2)

[Filtern](#)

Suchergebnisse

<< < 1

BRIEF OHNE UMSCHLAG

Karl Haeckel an Ernst Haeckel

ID: 35410
Verfasser: Haeckel,
Empfänger: Haeckel,
Datierung: 11.02.184

BRIEF OHNE UMSCHLAG

Karl Haeckel an Ernst Haeckel

ID: 35413
Verfasser: Haeckel,
Empfänger: Haeckel,
Datierung: 13.02.184

BRIEF OHNE UMSCHLAG

Karl Haeckel an Ernst Haeckel

ID: 35412
Verfasser: Haeckel,
Empfänger: Haeckel,
Datierung: 23.06.18

BRIEF OHNE UMSCHLAG

Karl Haeckel an Ernst Haeckel

ID: 35411
Verfasser: Haeckel,
Empfänger: Haeckel,
Datierung: 15.02.184

BRIEF OHNE UMSCHLAG

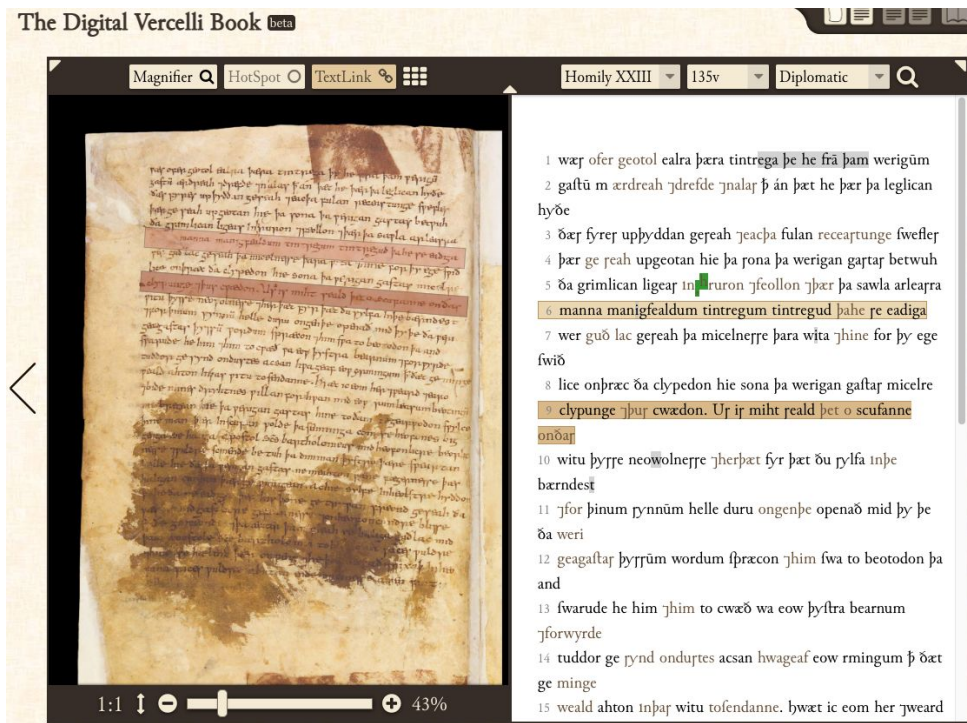
Karl Haeckel an Charlotte un

<https://haeckel-briefwechsel-projekt.uni-jena.de/en>

Text / Image Linking

The Digital Vercelli Book beta

Magnifier 🔍 HotSpot ○ TextLink 🔗 135v Homily XXIII Diplomatic



1 wæſ ofer geotol calra þæra tintrega þe he frā þam werigūm
2 gaſtū m ærdreah ȝdrefde ȝnalar þ̅ ān þæt he þær þa leglican
hyðe
3 ðær fyrer upȝyddan gereah ȝeaþa fulan receartunge ſweſter
4 þær ge ȝeah upgeotan hie þa ſona þa werigan gaſtaſ betwuh
5 ða grimlican ligeaſ in ȝuron ȝſcollon ȝhær þa sawla arleaſra
6 manna manigfealdum tintregum tintregud þa he re cadiga
7 wer guð lac gereah þa micelneſſe þara wita ȝhine for þy ege
ſwið
8 lice onþræc ða clypedon hie ſona þa werigan gaſtaſ micelre
9 clypunge ȝhær cwædon. Uf iſ miht reald þet o ſcufanne
onðar
10 witu þyſſe neowolneſſe ȝherþæt fyr þæt ðu ȝylfa inþe
bærndeſt
11 ȝfor þinum ȝynnūm helle duru ongenþe openað mid þy þe
ða weri
12 geagaſtaſ þyſſum wordum ſþræcon ȝhim ſwa to beoton þa
and
13 ſwarude he him ȝhim to cwæð wa eow þyſtra bearnum
ȝforwyrd
14 tuddor ge ȝynd onduptes acsan hwageaſ eow rmingum þ̅ ðæt
ge minge
15 weald ahton inþaſ witu toſendanne. þwæt ic eom her ȝweard

<http://vbd.humnet.unipi.it/beta/#135v>



Information Overview

Inv. 00002, Cologne, Papyrus Collection



2R

1422x2514 (png)



PInv. 00002r

1422x2514 (png)

 Institute of Archaeology at the University of Cologne. The metadata and publicly available images for this object are licensed under a Creative Commons Attribution 4.0 International License.

Persistent link to this object:

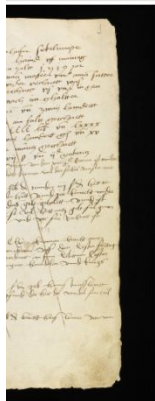
Trismegistus number	63496
Object type	role
material	papyrus
Dimensions	Width: 5.6 cm • Height: 10.2 cm
Texts	Recto : Adespotum: Prose fragment

bibliography	ILLUSTRATION	P.Köln III Panel Va
--------------	--------------	---------------------

Texts

Recto : Adespotum: Prose fragment			
Text type	literary	Dating	2nd century
Origin			
Trismegistus	63496	Permalink	https://papyri.uni-koeln.de/stueck/tm63496
Languages	Greek	Writing	Greek
Writing direction	parallel to the fibers	ink	Black

[le page](#) [Thumbnails](#)



Document Details

Annotations

Additional Bibliography (1)

Country of Location:

Location:

Library / Collection:

Shelfmark:

Manuscript Title:

Caption:

Language:

Manuscript Summary:

DOI (Digital Object Identifier):

Permanent link:

IIIF Manifest URL:

How to quote:

Online Since:

Rights:

Switzerland

Appenzell

Landesarchiv Appenzell Innerrhoden

LAAI, C.II.01

Book of accounts 1519-1520

Paper · 126 pp. · 31 x 24 cm · 1519-1520

German

On 126 paper pages, this volume contains the income and expenditures of the territory of Appenzell from February 1519 to February 1520. These accounts are among the most important sources for researching the history of the yet undivided region. (fre)

[10.5076/e-codices-laai-C-02-0001](https://dx.doi.org/10.5076/e-codices-laai-C-02-0001) (<http://dx.doi.org/10.5076/e-codices-laai-C-02-0001>)

<https://www.e-codices.unifr.ch/en/list/one/laai/C-02-0001>

 <https://www.e-codices.unifr.ch/metadata/iiif/laai-C-02-0001/manifest.json>

Appenzell, Landesarchiv Appenzell Innerrhoden, LAAI, C.II.01: Book of accounts 1519-1520 (<https://www.e-codices.unifr.ch/en/list/one/laai/C-02-0001>).

10/08/2020

Images:  (Concerning all other rights see each manuscript description and our Terms of use)

<https://papyri.uni-koeln.de/stueck/tm63496>

<https://e-codices.unifr.ch/en/list/one/laai/C-02-0001>

Registers / Indices

e-codices Person Index

Search:

All Author Commentator Translator Scribe Illuminator Bookbinder

A-Z A B C D E F G **H** I J K L M N

Quick Filter:

- Ha-ARI, Kitve
→ Luria, Yitshak ben Shelomoh (1534-1572) **Author**
- Ha-Darshan, Moses
→ HaDarshan, Moshe (XI s.) **Author**
- Ha-Tsarefati
→ Barukh ben Yitshak (1140-1212) **Author**
- HaDarshan, Moshe (XI s.) **Author**
- Hadasschan, Moscheh
→ HaDarshan, Moshe (XI s.) **Author**
- Haemmerlein, Felix
→ Hemmerlin, Felix (1389-1458) **Author**
- Haffner, Anton (-1608) **Author**
- Haffner, Franciscus
→ Haffner, Franz (1609-1671) **Author**
- Haffner, Frantz
→ Haffner, Franz (1609-1671) **Author**
- Haffner, Franz (1609-1671) **Author**
- Hai Ben Sherira, Gaon
→ Hai Ben Sherira, Ga'on (939-1038) **Author**

https://e-codices.unifr.ch/en/person_index



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PLACES

Home > Indices > Places

1363 Search Results

« previous 1 2 3 ... 137 next »



Aachen

50.77664 6.08342

Identifier: A130067



Aarau

47.39254 8.04422

Identifier: A130328



Aarburg

47.32067 7.89986

Identifier: A130329



Aachen

48.63115 8.07607

Identifier: A130917

<https://weber-gesamtausgabe.de/en/Indices/Places>

6.113 Results in the register "Plants & Fungi"

Search (e.g. 'Haller', 'Bern') **Plants & Fungi**

☒ Structural data ☒ Text contents

Results Open selection in new tab

☐	Entity group	Name	Type	Group of organisms	German name
☐	Plant / Fungus	Fraxinus excelsior (Nutzpflanze)	Crop	Flowering plant	Esche
☐	Plant / Fungus	Rhamnus cathartica (Nutzpflanze)	Crop	Flowering plant	Gemeiner Kreuzdorn
☐	Plant / Fungus	Ulmus (Nutzpflanze)	Crop	Flowering plant	Ulm
☐	Plant / Fungus	Rhamnus alpina (Nutzpflanze)	Crop	Flowering plant	Alpen-Kreuzdorn
☐	Plant / Fungus	Celtis australis (Nutzpflanze)	Crop	Flowering plant	Zürgelbaum

<https://hallernet.org/data/plants?query=&scope=mt&filters=>

Data and metadata in old-school print editions

252. Circular Letter to Strategoi

289a+b

5 × 6.9 cm (fr. 1), 4.1 × 4.9 cm (fr. 2)

212–17 or 222–35
Provenance unknown

Two fragments written across the fibres on the back of a document of which too little remains for its nature to be established. The first fragment preserves parts of the beginning of a letter to the strategoi of the Heptanomia and the Arsinoite nome from an official who was probably named Vitalis. This was clearly a copy. The second fragment is from the top of another column and probably contains more of the same text. It refers to Caracalla or Severus Alexander, which fixes the date to 212–17 or 222–35.

The letter concerns the publication of an edict in Alexandria, which may have been related to the crown tax; see fr. 1.5 n. Circulars to strategoi were usually issued by prefects, but what remains of the official's name cannot be matched to any known prefect, and it is difficult to place him in the *fasti* of this period. If he was not a prefect, he would have been a high-ranking procurator based in Alexandria. The absence of a title implies that he was not an acting prefect.



Fr. 1

Οὐ[τά]λις στρατηγοῖς Ἐ[πὶ τὰ νομῶν
καί] Ἀρσινόε[υ] [χαίρειν.
τοῦ προτε]θέντος διατάγμα[τος
πόλε]ι τὸν Ἀλεξαν[δρέων
], τεφ[.], . [

5



Fr. 2

]ν ἡ δὲ παροῦσα [
]τῆς πρὸς τὸν με[
]τορα καὶ κύριον [
Αὐρή]λιον Σεουήρον [
], . αρχία γενη[.], [

Can a computer understand and use that information to create interfaces or do analyses?

Data and metadata in old-school print editions

Some number (Inventory? Publication?)

Another number / ID

Dimensions per frag.

No. of frags.

Writing direction

Dating of sorts

No idea where it was from

Name of person (receiver)

Name of person (mentioned)

Name of place

252. Circular Letter to Strategē

289a+b

5 x 6.9 cm (fr. 1), 4.1 x 4.9 cm (fr. 2)

212-17 or 222-35
Provenance unknown

Two fragments written across the fibres on the back of a document of which too little remains for its nature to be established. The first fragment preserves parts of the beginning of a letter to the strategi of the Heptanomia and the Arsinoite nome from an official who was probably named Vitalis. This was clearly a copy. The second fragment is from the top of another column and probably contains more of the same text. It refers to Caracalla or Severus Alexander, which fixes the date to 212-17 or 222-35.

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Fr. 1

Fr. 2

Οὐ[]τάλις στρατηγοῖς Ἐ[πτά νομιῶν
καὶ Ἀρσινόει[σ]ο[] [χαίρειν.
τοῦ προτε]θέντος διατάγμα[τος
πάλε]ι τὸν Ἀλεξαν[δρέων
], τεφ[], .[]

5

Ἰν ἡ δὲ παρούσα []
[τῆς πρὸς τὸν με[]
[τοῖα καὶ κύριον []
Αὐρή[]λιον Σεουήρον []
], . αρχίας γενη[], .[]

Publication number: 252
Inventory number: 289a+b
Title: Circular Letter to Strategē
Type: Letter
No. of fragments: 2
Dimensions in cm (Fr. 1): 5 x 6.9
Dimensions in cm (Fr. 2): 4,1 x 4,9
Provenance: -
Dating hypothesis 1: 212-17
Dating hypothesis 2: 222-35
Name (receiver of letter): Vitalis
Name (mentioned): Caracalla
Name (mentioned): Severus Alexander
Place (mentioned): Alexandria

252 (Publication number)
 289a+b (Inventory number)
 Circular Letter to Strategē (Title)
 Letter (Type)
 2 (No. of fragments)
 5 x 6.9 (Dimensions in cm, Fr. 1)
 4,1 x 4,9 (Dimensions in cm, Fr. 2)
 - (Provenance)

...



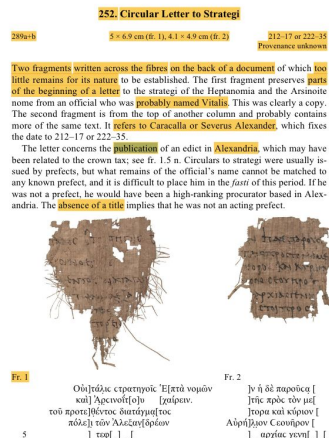
(structured) Data

Mentioned Places (for each):
Name: string of letters
Modern name (?): string of letters
Coordinates (?): string of letters

Data model (= blueprint)

= DATA MODELING

Data and metadata in old-school print editions



Object

Publication number: *some number*
Inventory number: *some alphanumeric number*
Title: *words*
Type: *either Letter or Receipt or Prose or Verses or ...*
No. of fragments: *a number*

For each **Fragment** we need:
Dimensions in cm: two numbers (with & height)
Provenance: name / id of a location
Dating hypothesis: a date (range)

For each **Text** we need:
Mentioned Names (for each):
 Who?: string of letters
 Role?: either Sender or Receiver or Mentioned

Mentioned Places (for each):
Name: string of letters
Modern name (?): string of letters
Coordinates (?): string of letters

Data model (= blueprint)

Publication number: 252
Inventory number: 289a+b
Title: Circular Letter to Strategion
Type: Letter
No. of fragments: 2
Dimensions in cm (Fr. 1): 5 x 6.9
Dimensions in cm (Fr. 2): 4,1 x 4,9
Provenance: –
Dating hypothesis 1: 212–17
Dating hypothesis 2: 222–35
Name (receiver of letter): Vitalis
Name (mentioned): Caracalla
Name (mentioned): Severus Alexander
Name (mentioned): Alexandria

(structured) Data

Applying the model / mapping / data entry with the data model as template

Modeling problems: Datings

244. Copy of a Private Letter

588a

15.1 × 9.3 cm

Early first century

The top of a letter from Diogenes to Eutychides, his 'brother'. It is called a copy in the heading; such copies are common in official correspondence, but not in

In print publications, we find datings such as:

- around 1350
- beginning of the 12th century
- mid-14th century
- between the 12th and 14th century
- not before 1492



Geschlossene Zeiträume (exemplarisch 8. Jh.)

Datumsangaben (zulässig)

8. Jh.	701 - 800
8. Jh. 1. V.	701 - 725
8. Jh. 2. V.	726 - 750
8. Jh. 3. V.	751 - 775
8. Jh. 4. V.	776 - 800
8. Jh. 1. D.	701 - 733
8. Jh. 2. D.	734 - 766
8. Jh. 3. D.	767 - 800

8. Jh. 1. H.	701 - 750
8. Jh. 2. H.	751 - 800
8. Jh. M.	741 - 760
um 750	745 - 755
8. Jh. A.	701 - 710
8. Jh. E.	791 - 800
um 800	795 - 805

Offene Zeiträume (exemplarisch 13. Jh.)

Datumsangaben (zulässig)

vor 1242	1192 - 1241
nach 1242	1243 - 1292
1242 oder früher	1192 - 1242
1242 oder später	1242 - 1292
um 1200 oder früher	1145 - 1205
um 1200 oder später	1195 - 1255
vor 13. Jh.	1001 - 1200
nach 13. Jh.	1301 - 1500
13. Jh. oder früher	1001 - 1300
13. Jh. oder später	1201 - 1500

Data types: (finally a) systematic overview!

- a. Digital Facsimiles
- b. Textual data/Transcription
- c. Editorial annotations
- d. Metadata (Ask)
 - i. Administrative (bibliographic metadata, rights, provenance?)
 - ii. Descriptive (Layout, pages, writing support, content, codicology and palaeography)
 - iii. Structural (specific for digitized manuscripts)
 - iv. Technical Metadata

Metadata

“Data about data”

Administrative Metadata

- the bibliographic record of the digitized object
- common standards
 - MARC standards
 - Library of Congress
 - MACHine-REadable Cataloguing standard
 - Dublin Core Schema
 - Dublin Core Metadata Terms
 - vocabulary to describe digital object/resources
- Tracks the lifecycle of the digitized source

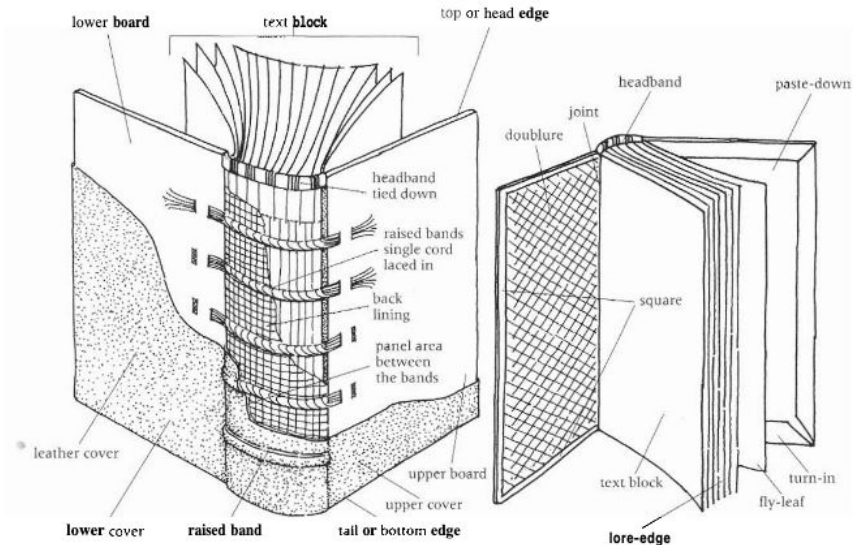
Descriptive Metadata

- semantic description of the digitized object
- less standardized
- specific for historical written documents
 - manuscripts are not 'born' with descriptive metadata
 - e.g. authorship, printing press, publisher, year of publication, contents
- ideally replaces the analogue catalogue
- should always be connected to the digital facsimile

Structural Metadata

- a book is held together by leaves, page numbers, quires, book block, and book cover
- structural metadata describes and connects the structure of a book
- keeps images in the right order
- connection to the bibliographical record

→ accommodates bibliographic and descriptive metadata
→ linking between paratextual elements
→ ensures a sustainable digital representation



source: <https://senna.sjsu.edu/lmain/medwriting/illus.pdf>

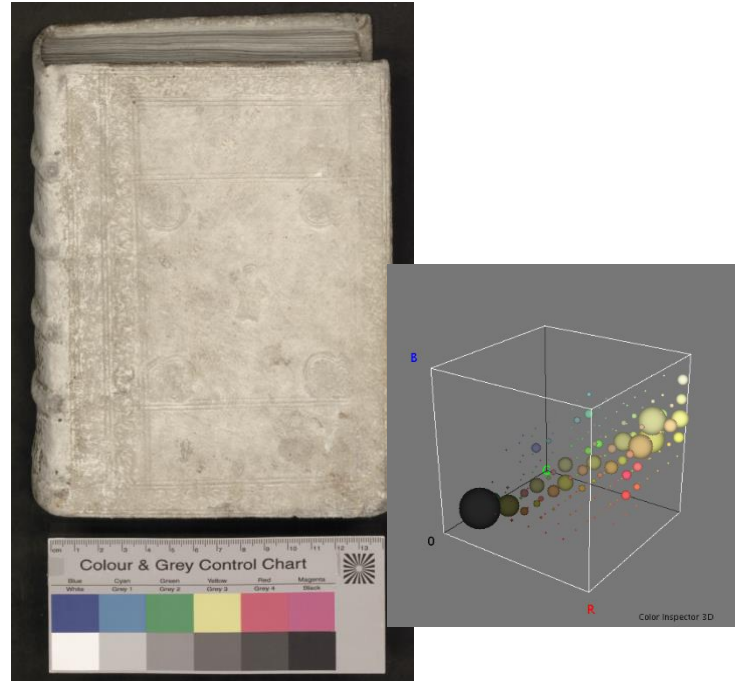
Technical Metadata

- file format
- resolution (dpi)
- bit depth
- colour/hue
- illumination
- information about digitization hardware (scanner/camera)

→ information about quality and reliability of the digital facsimile

→ important for post- and preprocessing (e.g. image analysis, ATR)

Example: Colour calibration

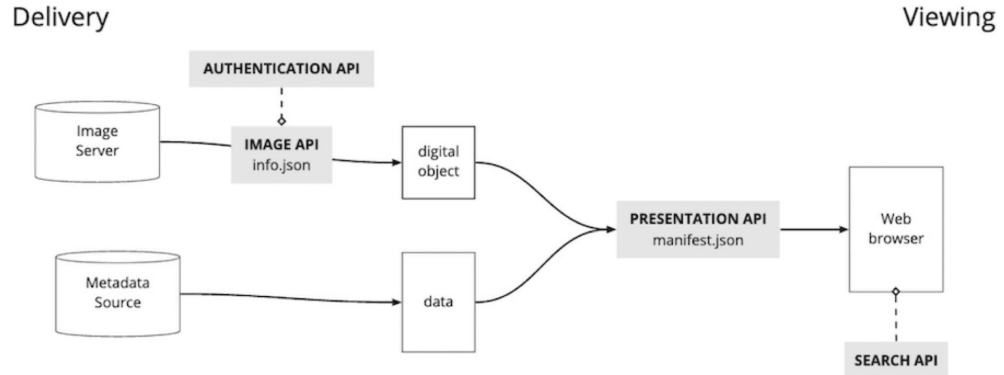




International Image Interoperability Framework

- scheme to facilitate better access to digital images at cultural heritage institutions
- services developed and maintained by a community network
- exchange of data between different computers and servers
- IIIF manifest (JSON file) contains image and metadata

→ Image API and Presentation API



About this item

CURRENT ITEM

4r

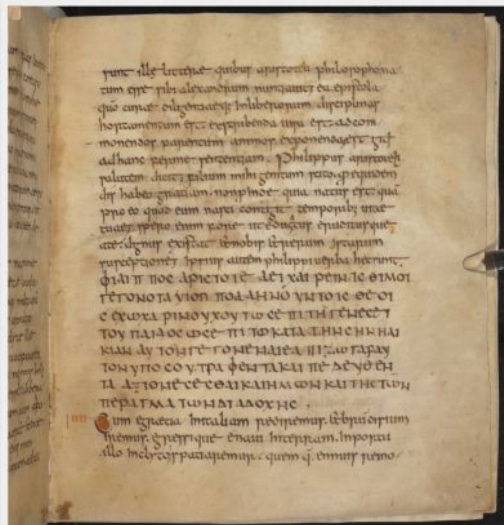
COLLECTION

SHOW COLLECTION

RESOURCE

Leeuwarden, Tresoar: PBF 55 hs

This almost square manuscript, written in 836 in Fulda (Germany), contains a copy of Aulus Gellius' *Noctes Atticae*. It is partly written in insular script, partly in Caroline minuscule in multiple hands. The manuscript has been identified with the one mentioned by Lupus of Ferrières in a letter: the date and localisation are based on that witness. Notes are found in the margins.



About this item

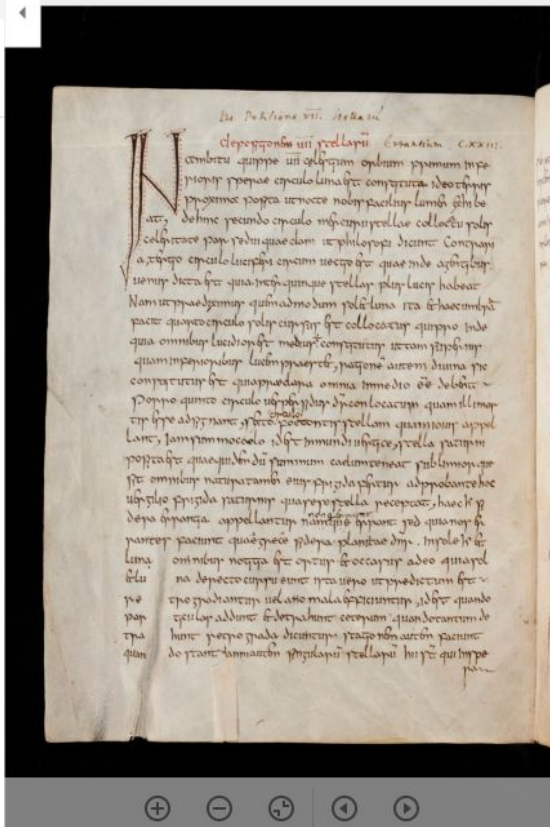
CURRENT ITEM

4v

RESOURCE

Basel, Universitätsbibliothek, F III 15a

One of the Isidore codices from the Monastery of Fulda; the codex escaped destruction because it reached Basel during the 16th century, before the abduction and destruction of the library during the Thirty Years' War. There it apparently was to serve as a textual source for a planned edition of Isidore's works. This codex was created in Fulda at the end of the 9th century and still retains its Carolingian binding in a parchment cover. In addition to the works of Isidore, it



Working with Digitized Manuscripts



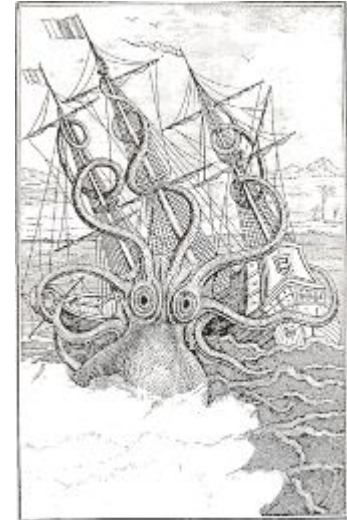
ATR, OCR, HTR - How Computers Transcribe Texts

Automatic Text Recognition

Optical Character Recognition

Handwritten Text Recognition

What's the difference?



Optical Character Recognition (OCR)

- optical scanning
- pattern recognition
- character based recognition
- line level

How to recognize OCR software?

- open source software usually named after cephalopods
- OCRopus, Kraken, Calamari

Handwritten Text Recognition

- transcription of handwritten text
- involves a certain degree of interpretation
- intelligent word recognition

Examples of HTR software

- Transkribus
- Loghi

Automatic Text Recognition

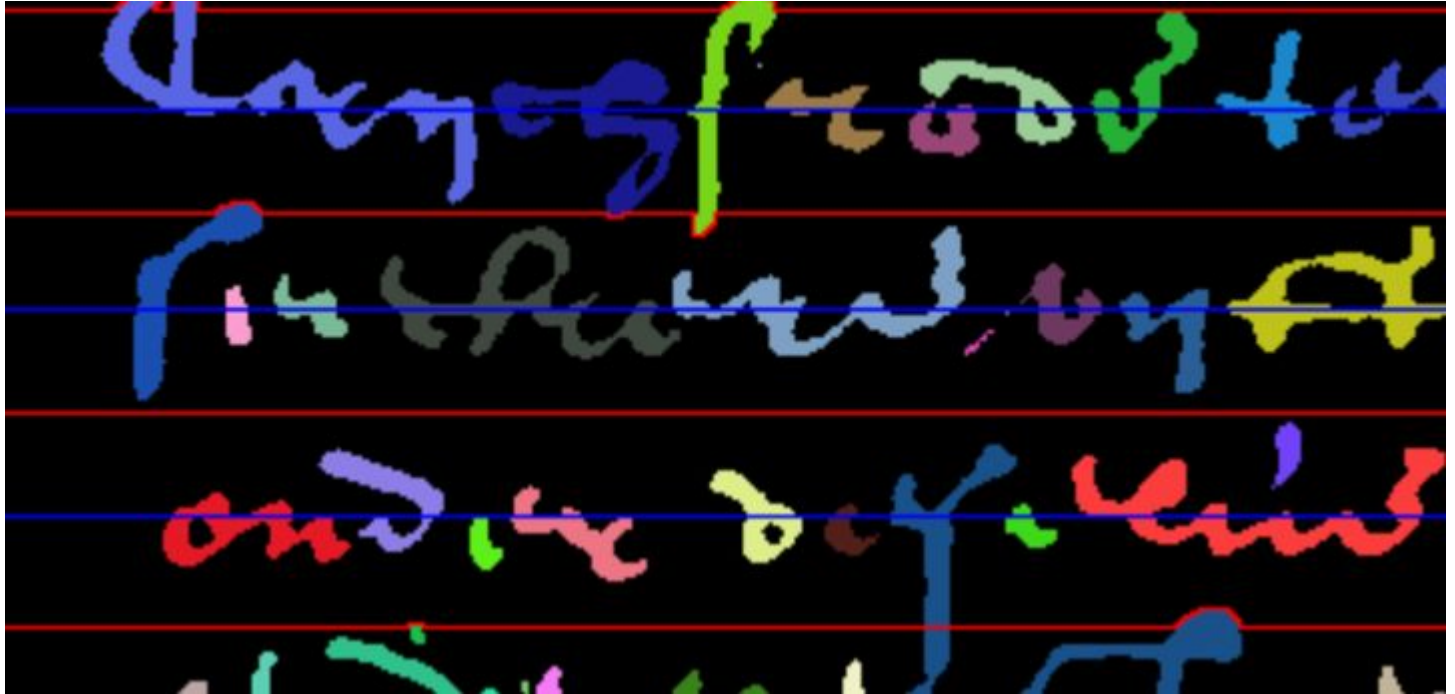
- the distinction of OCR and HTR has vanished with AI/ML applications
- many applications combine both
- sometimes OCR based software is more suited for manuscripts
- Large Language Models (LLMs) will impact ATR

→ pre-processing steps are always the same!

Layout analysis and segmentation



Line segmentation



q̄ n̄ ad penā uidet̄ d̄s s; ad gaudiū sempit̄nū. & n̄ intelligebat cū q̄ loqueret̄. q̄ tantū m̄ cū filiū

Venerantes memoria meius. cum sobrietate et preci

suū & matrē & uxōrē & filios & fr̄s & sorores. adhuc & animā

nescim̄. cōpulsi loco doctrine loquendo discim̄. & dū de

inētuitū cogitantis. quatuor species reperiunt̄. quasi gradatū nat̄. altera ex altera. se

terrenis cogitationib⁹ mutis & uariis. ecce uide si potes. D̄s ueritas ē. Hoc enī scriptū ē. quō d̄s lux ē.

n̄ eos sic & iā c̄sequentia docent anglos fuisse fateantur. An q̄ n̄ dictū ē angl̄s ei locut⁹ ē t ap



Dating and localizing medieval Latin manuscripts with the support of IIF and Machine Learning

displea
miserie
dialisee

RAMEN
iquapet

Capitula ma
femur adalond
aremdanetam

fur m fur se
nam legumur l

catis terram
Intellegite
ritales autem

unum in die ill
Panem nr̄m co
ctue stimenti

asnaribus ex
ocrates quia
r infans. Nec

pe apit angu
scriptura ill
ne paine ac t

us ligat

configit² spina
non opui. D
am dno. r tu

quos pueri quomqz ne
noy. Reliquas
ux eto opm̄ cedus gl
in plena potentia doleat
magistratus me dec
uata potius unibz tato

2 pongs no pubrix
fursumqz ad p
p̄ pletu ylu dig
lond tua .i. inga

is. p. q. l. si quis hac. q. iudiciu
mirenti tibi. q. lex illa. quē
opis etis. dū in unctis loqē
tū. l. illa. cum pa. q. illa si q
differunt in solutē. ēte em
possūt. in cē n. ut ff. ope un
cat. in illa. L. ope oficiales o
it in cē fabriles uō etas. qu

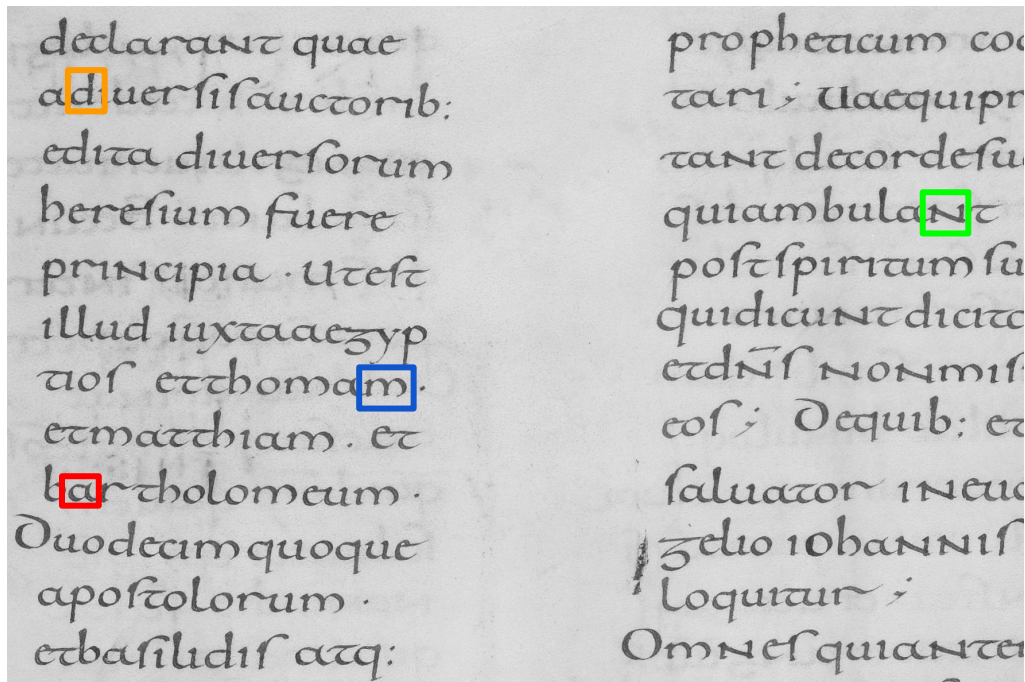
patris
facient

le lemp uny
marie gent
dei et domini

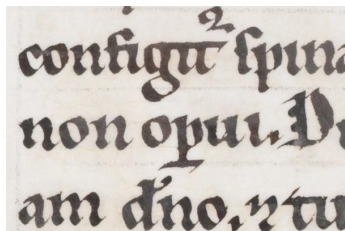
Black Box Palaeography

Half-Uncial

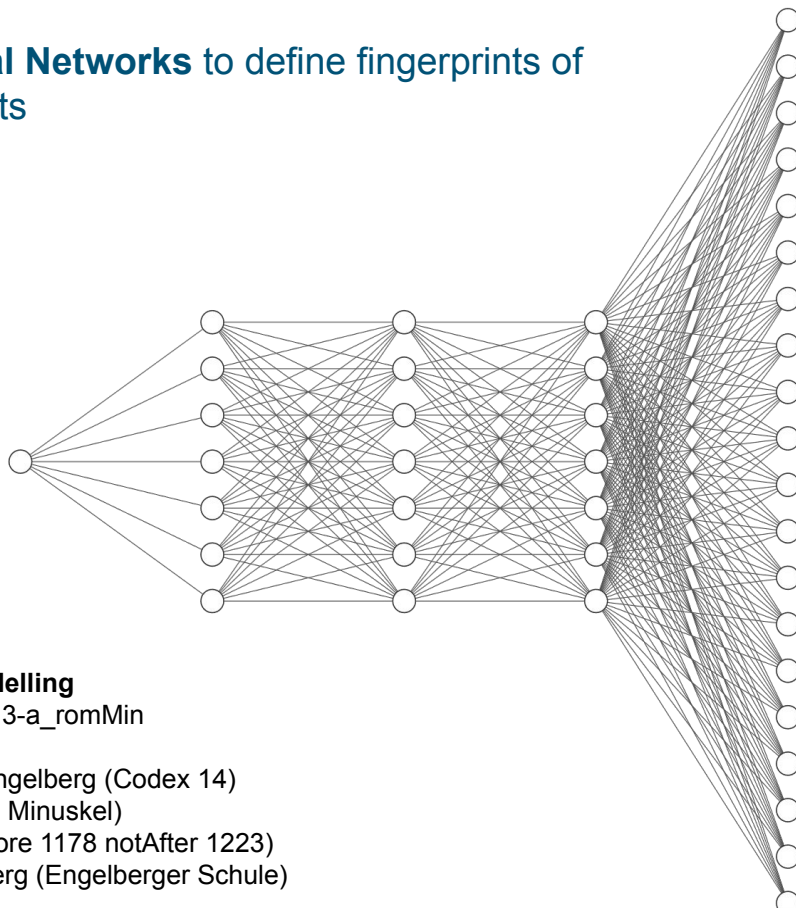
- Minuscule script (ascenders and descenders clearly between the lines)
- mainly open letter **a**; vertical **d**, **3** or **z**-shaped letter **g**; minuscule form of **m** (three minims); both **N** and **n** may be used; mostly long **s** (I = U+017F); **t** with loop on the left;
- The corpus encompasses more or less cursive types of half-uncial scripts



Testing **Deep Neural Networks** to define fingerprints of medieval book scripts



configit² spina
non opui. Dixi
am dno. et tu



Digital Forensics data modelling

ID-Dataset: 13_Engelberg_13-a_romMin

Labels:

Collection: Stiftsbibliothek Engelberg (Codex 14)

Script: romMin (Romanische Minuskel)

Date of Origin: 1197 (notBefore 1178 notAfter 1223)

Place of Origin: Stift Engelberg (Engelberger Schule)

Repository: eCodices



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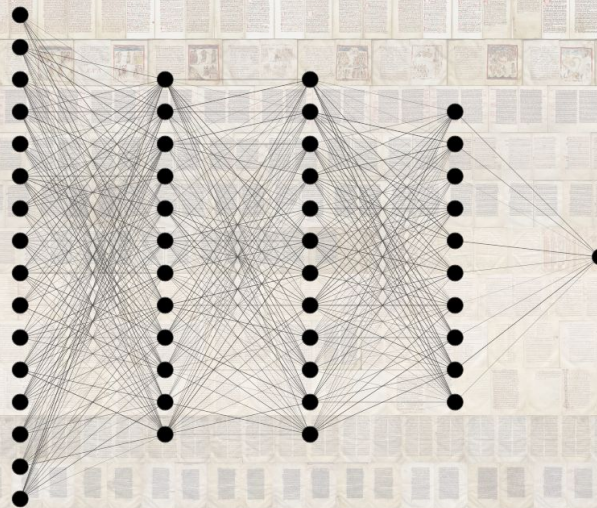
oppla. et oculi ei² gratia. Veni² in
intelligit² ex eo² psona dei. Quō die
ip me manus tua. Die fortasse cū
ceret nec ab eis cognosceret. nocte
eret. quando cernus putauerunt
pemptū & ex tunc. S; quō cū tra
erent² p²re²... q² oport
re cōpunchi. S; quō cū tra
in capite ei²... quoz re
ioz tecla s;... cū dictū e
tua tamq²... ut ian
viderent² poster²... a. sequit² uo
ut ex fide re²... pecc
tis. conuersus sum inquit in eru
spina. Peccatū meū cognoui. & u
ui. Dixi pnunciabo aduersū me
o. et tu dimisisti impietate cordis
arnis nubilo debem² inuolui. ut p
fui eē inuisibile. dorsū & uisibili
a serui utrūq; uisibilit² apparuit
ut p²...

Matching Medieval Manuscripts



1

2



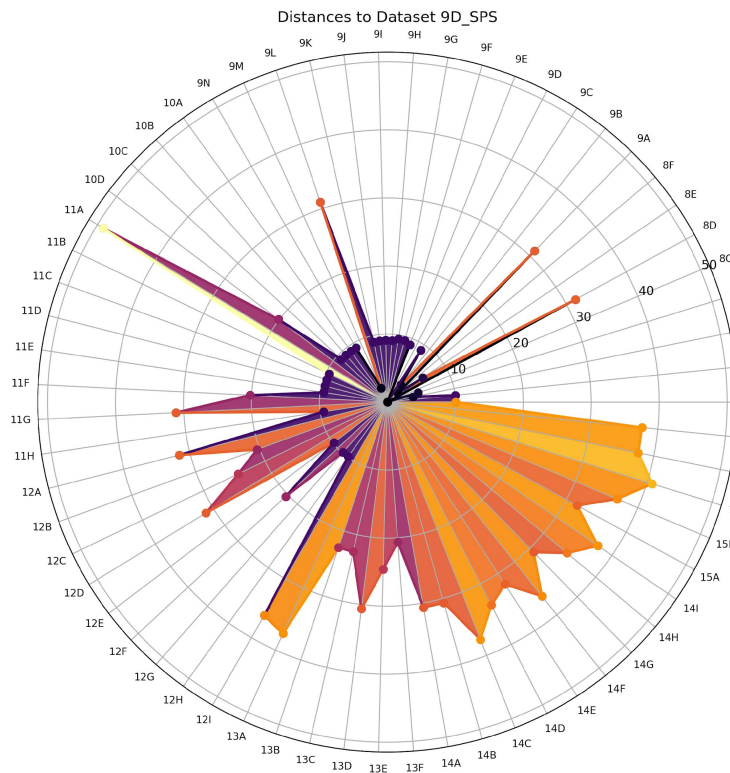
Is it a Match?

How similar are the script samples?



Learning by comparison with Conjoint Twin ConvNets

Matching Burgerbibliothek Bern, Codex 207 with 62 (SPS) and 55 (NN) manuscripts.



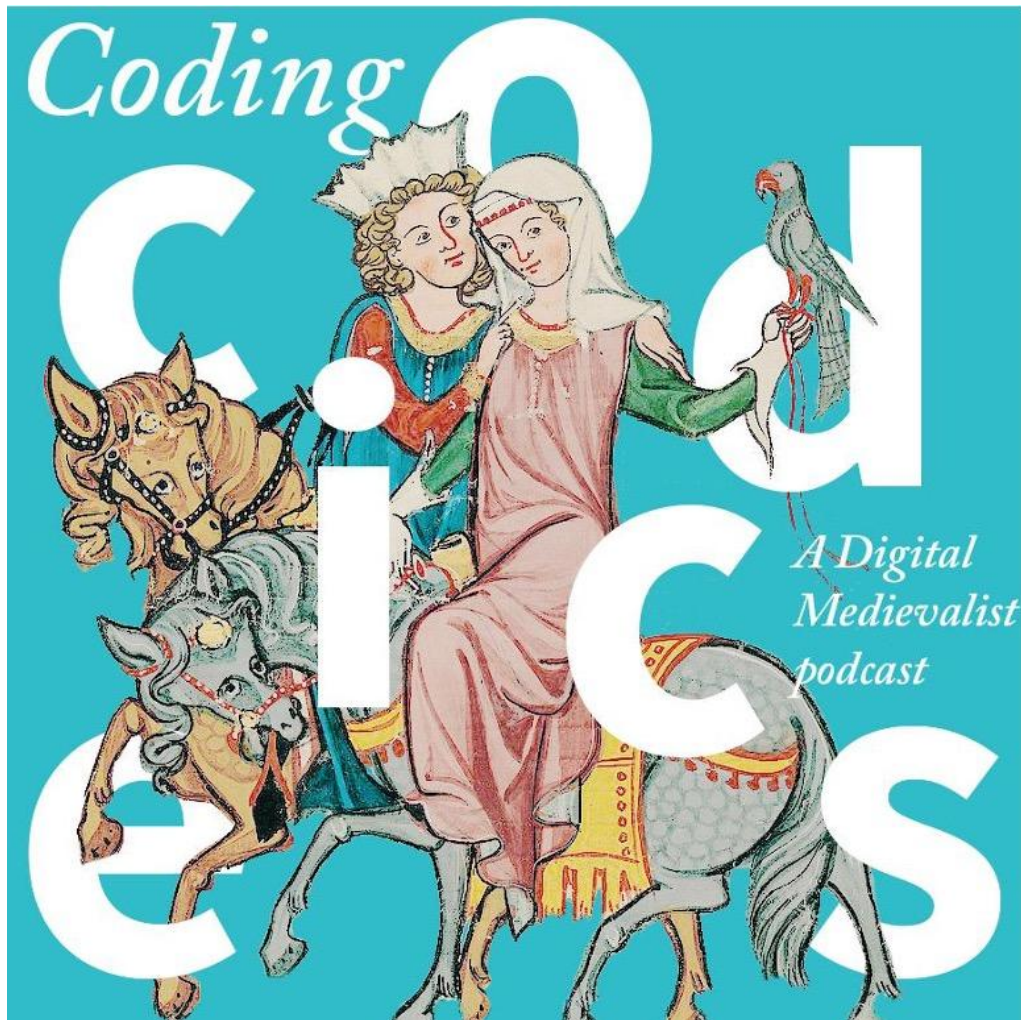
(dm/)

You want to listen more about Digital
Medieval Studies? Go to:

<https://codingcodices.wordpress.com/>



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Thank you!

Contact: @cesare_blanc@fedihum.org
hannah.busch@uni-koeln.de



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